Report Status FINAL
Route 2006 Ordered by:
Ironwood Cancer and Research Center Scottsdale
8880 E Desert Cove Ave
Scottsdale, AZ 85260

Sonora Quest Laboratories®

Rakesh Bagai, MD

Patient Information:
MILLER, ERIC J
Order #: 1505874172541241718 / NL124127782

Account: 15058
ID/MR#: 1759356
Patient Lab ID: 64af3bc5e50bb0441e44418d

Collected: 05/26/2026 08:56 AM
Received: 05/26/2026 08:31 PM
Reported: 05/30/2026 01:22 PM

DOB: 07/22/1967 Age: 58Y-10M-4D
Sex: M Fasting: No
Patient Phone: 480-287-2227

HEMATOLOGY

CBC w/ Differential, w/ Platelet

TEST	RESULTS	REFERENCE RANGES	UNITS	PL
WBC	8.0	4.0 - 11.0	k/mm3	
RBC	4.82	4.30 - 6.00	m/mm3	
Hemoglobin	16.5	13.5 - 17.0	g/dL	
Hematocrit	48.8	40.0 - 53.0	%	
MCV	**101.2 H**	78.0 - 100.0	fL	
MCH	**34.2 H**	27.0 - 34.0	pg	
MCHC	33.8	31.0 - 37.0	g/dL	
Platelet Count	257	130 - 450	k/mm3	
RDW(sd)	46.2	38.0 - 49.0	fL	
RDW(cv)	12.2	11.0 - 15.0	%	
MPV	10.5	9.0 - 12.0	fL	
Segmented Neutrophils	65.7*		%	
Lymphocytes	22.4		%	
Monocytes	10.8		%	
Eosinophils	0.5		%	
Basophils	0.2		%	
Absolute Neutrophil	5.3	1.5 - 7.8	k/uL	
Absolute Lymphocyte	1.8	0.9 - 3.9	k/uL	
Absolute Monocyte	0.9	0.2 - 1.0	k/uL	
Absolute Eosinophil	0.0	0.0 - 0.6	k/uL	
Absolute Basophil	0.0	0.0 - 0.2	k/uL	
Immature Granulocytes	0.4		%	
Absolute Immature Granulocytes	0.0	0.0 - 0.1	k/uL	
NRBC RE, Nucleated Red Blood Cell Percent	0.0	0.0 - 1.0	%	

*Segmented Neutrophils: Automated Diff

CHEMISTRY

Specimen Integrity

Slightly hemolyzed specimen: Potassium results may be increased due to erythrocyte contamination or hemolysis.

TEST	RESULTS	REFERENCE RANGES	UNITS	PL
Protein, Total	7.0	6.0 - 7.7	g/dL	
Beta-2-Microglobulin	1.8	0.8 - 2.4	mg/L	

Values obtained with different assay methods or kits cannot be used interchangeably. If this has occurred, re-baseline testing may be necessary. This test was performed using the Roche Cobas Tina-Quant B2MG immunoturbidometric assay.

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Comprehensive Metabolic Panel

Glucose	74*	70 - 99	mg/dL
Urea Nitrogen (BUN)	8	7 - 28	mg/dL
Creatinine	0.70	0.68 - 1.37	mg/dL
eGFRcr CKD-EPI	106*	≥60	mL/min/1.73m2
BUN/Creatinine Ratio	11.4	10.0 - 28.0	
Sodium	139	135 - 145	mmol/L
Potassium	4.4	3.6 - 5.3	mmol/L
Chloride	**97 L**	98 - 108	mmol/L
Carbon Dioxide (CO2)	25	20 - 31	mmol/L
Anion Gap	17	4 - 18	
Albumin	4.8	3.8 - 5.1	g/dL
Globulin	2.2	1.7 - 3.3	g/dL
Albumin/Globulin Ratio	2.2	1.3 - 2.7	
Calcium	10.4	8.7 - 10.4	mg/dL
Alkaline Phosphatase	66	40 - 140	IU/L
Alanine Aminotransferase	31	5 - 60	IU/L
Aspartate Aminotransferase	24	12 - 47	IU/L
Bilirubin, Total	0.3	≤1.2	mg/dL

*Glucose: Glucose reference range reflects fasting state.
*eGFRcr CKD-EPI: eGFRcr calculated using the CKD-EPI 2021 equation

Immunotyping (IT) w/ IgG, IgA, IgM Panel

IgG	**629 L**	694 - 1618	mg/dL
IgA	**39 L**	81 - 463	mg/dL
IgM	**26 L**	48 - 271	mg/dL
IT Interpretation	See Comment*		

*IT Interpretation: Two monoclonal gammopathies, IgG, kappa type; duplication often represents the presence of a structural variant of an individual immunoglobulin clone.

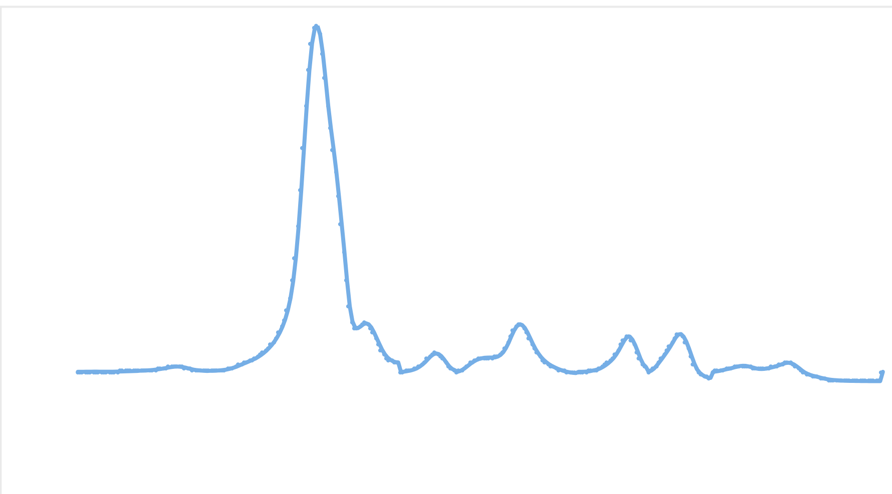
Protein Electrophoresis, Serum

Serum Proteins	Graph		
Albumin	4.49	3.52 - 5.29	g/dL
Alpha-1-Globulin	0.28	0.18 - 0.39	g/dL
Alpha-2-Globulin	0.91	0.45 - 0.94	g/dL
Beta Globulin 1	0.44	0.30 - 0.58	g/dL
Beta Globulin 2	0.50	0.20 - 0.52	g/dL
Gamma Globulin	**0.37 L**	0.70 - 1.50	g/dL
A/G Ratio	1.80		
M-Spike 1	**0.36 H**	≤0.00	g/dL
M-Spike 2	**0.13 H**	≤0.00	g/dL
PE Interpretation	See Comment*		

*PE Interpretation: Monoclonal gammopathy in the beta region; compatible with plasma cell dyscrasia. Suggest serum immunotyping to verify findings and determine protein isotype.

Monoclonal band co-migrates with the beta globulin. Quantitation is performed but may be affected. Recommend use of total immunoglobulin measurements (IgG, IgA, or IgM) to monitor changes.

Monoclonal gammopathy in the gamma region; compatible with plasma cell dyscrasia. Suggest serum immunotyping to verify findings and determine protein isotype.



Free Kappa/Lambda L/C Ratio,Serum

Free Kappa Light Chain	**64.74 H**	3.30 - 19.40	mg/L
Free Lambda Light Chain	**3.46 L**	5.71 - 26.30	mg/L
Free Kappa/Lambda Ratio	**18.71 H ***	0.26 - 1.65	

*Free Kappa/Lambda Ratio: Interpretation: In serum, the kappa/lambda ratio of whole immunoglobulin molecules is 2:1, whereas the kappa/lambda ratio of free light chains is 1:1.5. The latter is attributed to the occurrence of lambda light chains as dimers whose serum half-life is approximately 3 times longer than that of monomeric kappa light chains. Excess production of kappa or lambda light chains alters the kappa/lambda ratio. Alterations that fall outside of the normal range are attributed to the presence of monoclonal light chains. Monoclonal light chains are found in serum of patients with multiple myeloma, the light chain variant of MM, Waldenstrom's macroglobulinemia, mu-heavy chain disease, primary amyloidosis, light chain deposition disease, monoclonal gammopathy of undetermined significance, and lymphoproliferative disease such as B-CLL. Measurement of free light chain concentration in serum is useful for diagnosis, prognosis, monitoring disease activity and following response to therapy of these disorders.

Chronic infection and chronic inflammatory diseases, as well as renal insufficiency, may be accompanied by a diffuse increase in both kappa and lambda free light chains, but the kappa/lambda ratio remains within the normal limits. The serum concentration of free light chains increases with age over 60 years; light chains may reach 50 mg/L in those 70-80 yrs of age; in these cases the kappa/lambda ratio still remains within normal limits.

Physicians, who are accustomed to the identification of clonal protein by electrophoretic means, may order immunofixation in addition to free light chain immunoassay. In rare instances, immunofixation may identify monoclonal light chain protein in the absence of abnormalities in the quantitative light chain immunoassays.

Katzman JA et al., Serum reference intervals and diagnostic ranges for free kappa and free lambda immunoglobulin light chains: Relative sensitivity for detection of monoclonal light chains. Clin Chem 2002,48:1437-1444.

Tests Ordered: CBC w/ Differential, w/ Platelet; Beta-2-Microglobulin; Comprehensive Metabolic Panel; Free Kappa/Lambda L/C Ratio,Serum; Monoclonal Gammopathy Monitoring Panel (SPE, IT)

Values Outside of Reference Range

TEST	RESULTS	REFERENCE RANGES	UNITS
MCV	**101.2 H**	78.0 - 100.0	fL
MCH	**34.2 H**	27.0 - 34.0	pg
Chloride	**97 L**	98 - 108	mmol/L
IgG	**629 L**	694 - 1618	mg/dL
IgA	**39 L**	81 - 463	mg/dL
IgM	**26 L**	48 - 271	mg/dL
Gamma Globulin	**0.37 L**	0.70 - 1.50	g/dL
M-Spike 1	**0.36 H**	≤0.00	g/dL
M-Spike 2	**0.13 H**	≤0.00	g/dL
Free Kappa Light Chain	**64.74 H**	3.30 - 19.40	mg/L
Free Lambda Light Chain	**3.46 L**	5.71 - 26.30	mg/L
Free Kappa/Lambda Ratio	**18.71 H**	0.26 - 1.65	

Values listed above may not include all results considered abnormal for this patient (e.g., text-only results, such as those for some pathology/cytology specimens, and results for analytes without established reference ranges will not appear). Always review the entire patient report and correlate all results with the patient's clinical condition.

Unless otherwise noted, testing performed by: Sonora Quest Laboratories, 424 S 56th St, Phoenix, AZ 85034 800.766.6721
Laboratory Director: Dr. Timothy McDonald

End of Report